A book chapter in Digital Multimedia Perception

Statistical Models and Algorithms for Image Annotation and Retrieval: Mathematics of Perception

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1. Introduction

The principal design goal of a visual information retrieval system is to return data (images or video clips) that accurately match users' queries (for example, a search for pictures of a deer). To achieve this design goal, the system must first comprehend a user's query concept (i.e., a user's perception) thoroughly, and then find data in the low-level input space (formed by a set of perceptual features) that match the concept accurately [1]. Statistical learning techniques can assist achieving the design goal via two complementary avenues: *semantic annotation* and *query-concept learning*.

Semantic annotation provides visual data with semantic labels to support keyword-based searches (for example, landscape, sunset, animals, and so forth). Several researchers have proposed semi-automatic annotation methods to propagate keywords from a small set of annotated images to the other images [2,3,4,5]. Although semantic annotation can provide some relevant query results, annotation is often subjective and narrowly construed. When it is, query performance may be compromised. To thoroughly understand a query concept, with all of its semantics and subjectivity, a system must obtain the target concept from the user directly via *query-concept learning* [6,7]. *Semantic annotation* can assist, but not replace, *query-concept learning*.

Both *semantic annotation* and *query-concept learning* can be cast into the form of a supervised learning problem [8,9], which consists of three steps. First, a representative set of perceptual features (e.g., color, texture and shape) is extracted from each training instance. Second, each training feature-vector $\mathbf{x}_i$ is assigned semantic labels g_i. Third, a classifier $f(.)$ is trained by a supervised learning algorithm, based on the labeled instances, to predict the class labels of a query instance $\mathbf{x}_q$. Given a query instance $\mathbf{x}_q$ represented by its low-level features, the semantic labels g_q of $\mathbf{x}_q$ can be predicted by $g_q = f(\mathbf{x}_q)$. In essence, these steps learn a mapping between the perceptual features and a human perceived concept.

At first it might seem that traditional supervised learning methods such as neural networks, decision trees, and Support Vector Machines could be directly applied to perform *semantic annotation* and *query-concept learning*. Unfortunately, such traditional learning algorithms are not adequate to deal with the technical challenges posed by these two tasks. To illustrate, let D denote the number of low-level features, N the number of training instances, N^+ the number of positive training instances, and N^- the number of negative training instances ($N = N^+ + N^-$). Two major technical challenges arise:

1. Scarcity of training data. The features-to-semantics mapping problem often comes up against the D > N challenge. For instance, in the query-concept learning scenario, the number of low-level features that characterize an image (D) is greater than the number of images a user would be willing to label (N) during a relevance feedback session. As pointed out by Donoho

[10], the theories underlying "classical" data analysis are based on the assumptions that D < N, and N approaches infinity. But when D > N, the basic methodology which was used in the classical situation is not similarly applicable.

2. Imbalance of training classes. The target class in the training pool is typically outnumbered by the non-target classes ($N^- >> N^+$). For instance, in a k-class classification problem where each class has about the same number of training instances, the target class is outnumbered by the non-target classes by a ratio of k-1:1. The class boundary of imbalanced training classes tends to skew toward the target class when k is large. This skew makes class prediction less reliable.

We use an example to explain the above technical challenges. Figures 1-3 show an example query using a *Perception-based Image Retrieval* (PBIR) [44] prototype developed at UC Santa Barbara on a *300,000*-image dataset. This dataset had been manually annotated by professionals. The prototype extracted perceptual features (described in the next section) from the images, using the PBIR learning methods to combine keywords and perceptual features in a synergistic way to support retrieval. The figures demonstrate how a query concept is learned in an iterative process by PBIR search engine to improve search results. The user interface shows two frames. The frame on the left-hand side is the feedback frame, on which the user marks images relevant to his or her query concept. On the right-hand side, the search engine returns what it interprets as matching this far from the image database.

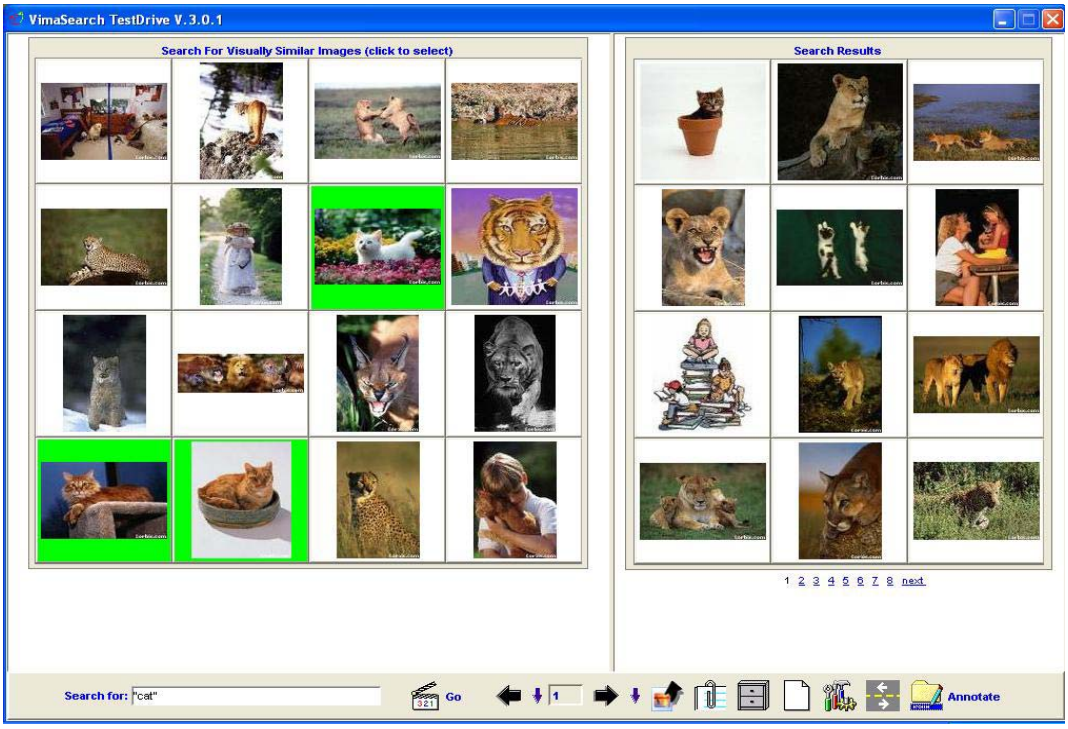


Figure 1. Cat Query Initial Screen.

To query "cat," we first enter the keyword *cat* in the query box to get the first screen of results in Figure 1. The right-side frame shows a couple of images containing domestic cats, but several images containing tigers or lions. This is because many tiger/lion images were annotated with "wild cat" or "cat." To disambiguate the concept, we click on a couple of domestic cat images on the feedback frame (left side, in gray borders). The search engine refines the class boundary

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accordingly, and then returns the second screen in Figure 2. In this figure, we can see that the images in the result frame (right side) have been much improved. All returned images contain a domestic cat or two. After we perform another couple of rounds of feedback to make some further refinements, we obtain the more satisfactory results shown in Figure 3.

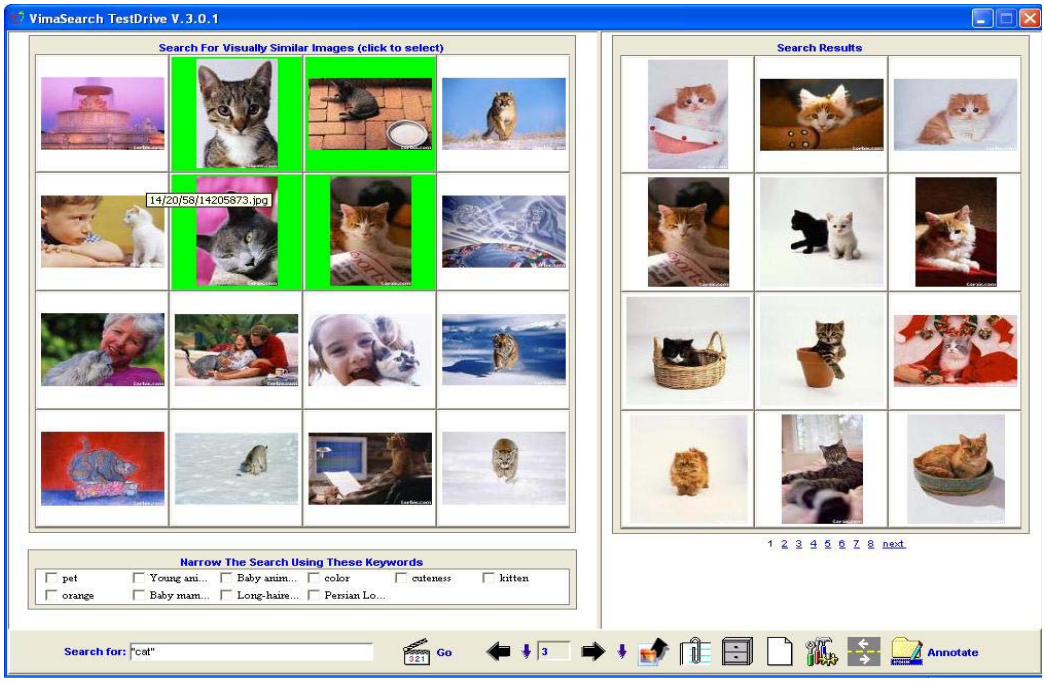


Figure 2. Cat Query After Iteration #2.

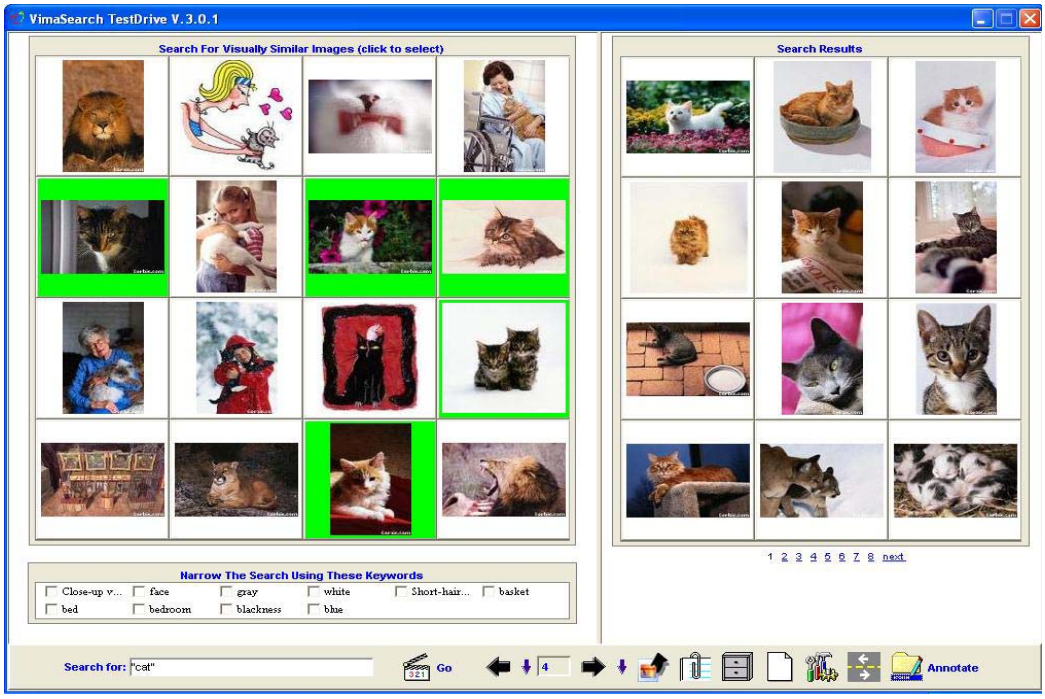


Figure 3. Cat Query After Iteration #3

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This example illustrates three critical points. First, keywords alone cannot retrieve images effectively because words may have varied meanings or senses. This is called the *word-aliasing* problem. Second, the number of labeled instances that can be collected from a user is limited. Through three feedback iterations, we can gather just $16 \times 3 = 48$ training instances, whereas the feature dimension of this dataset is *144* [7]. Since most users would not be willing to give more than three iterations of feedback, we encounter the problem of scarcity of training data. Third, the negatives outnumber the relevant or positive instances being clicked on. This is known as the problem of imbalanced training data.

In the remaining sections of this article, we present our proposed statistical methods for dealing with the above challenges for mapping perceptual features to perception. Section 2 depicts the perceptual features we use to characterize images. Section 3 presents the training-data scarcity problem in more detail and outlines three proposed remedies: *active learning*, *recursive subspace co-training*, and *adaptive dimensionality reduction*. Section 4 discusses the problem of training-class imbalance and outlines two remedies, *class-boundary alignment* and *quasi-bagging*. Section 5 sketches EXTENT, a proposed architecture for combining context and content (perceptual features) to annotate images, and CDAL, an active learning algorithm that considers both keywords and perceptual features.

2. Perceptual Features

The first step to model perception is to extract features from images that can represent how the human visual process perceives an image [51]. Our vision tends to pay less attention to the boarder pixels, and it distributes effort unevenly by paying closer attention to ambiguous regions. Our visual system adjusts to the environment and adapts to the visual goals. For instance, we may not be able to tell if a figure is a real person or a statue at first glance. If we pay closer attention to the figure's surface (texture), we may be able to identify it as a statue or person. We thus believe that an image search engine must also be adaptive to the goals of a search task. We can think of our visual system as being divided into two parts: our eyes (the front-end) perceive images and our brain (the back-end that is equipped with a knowledge database and an inference engine) recognizes images. The front-end collects visual data for the back-end to allow high-level processing. The back-end instructs the front-end to zoom, pan, and collect visual data with different features. (A feature can be regarded as a particular way of perceiving an image.) The front-end responds flexibly in perceiving visual data by selecting, ordering and weighting visual features differently. These features can be categorized into *holistic* and *local* perceptual features.

Holistic perceptual feature. For each image, we extract global features, which include *color*, *texture*, *shape*, etc. [53. We divide pixels on an image into color sets. A color set is the collection of pixels in an image that have the same color. We divide the visible color space into eleven culture colors [7]. An image thus consists of at most eleven color sets, where a color set is characterized by three attributes of the image: its color, moments and centroid. These attributes are defined as follows:

- Color: The culture color of the pixel collection. Each culture color spans a range of wavelengths. We keep the mean and variance of the color set on the three axes: hue, saturation, and intensity. We also record the number of pixels belonging to the color set.

- Moments: The moment invariants [52] of the color set. Moment invariants are properties of connected regions in binary images that are invariant to translation, rotation and scaling.

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- Centroid: The center of gravity of the color set in x and y.

We also extract texture features in three orientations (vertical, horizontal, and diagonal) and three resolutions (coarse, medium, and fine). Both color and texture are characterized in multiple resolutions. Shapes are treated as attributes of color and texture and characterized by moments. (For details, please refer to [7].)

Local perceptual features. Scale Invariant Feature Transform (SIFT) has become a very popular local feature representation in recent years [38,39]. SIFT extracts local, distinctive, invariant features. SIFT consists of four major stages: 1) scale-space extreme detection, 2) keypoint localization, 3) orientation assignment, and 4) keypoint descriptor. Features generated by SIFT have been shown to be robust in matching across an affined distortion, change in 3D viewpoint, addition of noise, and change of illumination.

The key of our features is that they represent images in multiple scales, orientation, and aspects. These features will then be combined in a non-linear way to formulate a query concept.

3. Scarcity of Training Data

Some of the first image retrieval systems required a sample picture from the user to begin a search (e.g., [45,46,47,48]). Yet, the limitations of that procedure quickly become obvious. A user, for example, might not have a picture of panda when he or she begins to seek one. Even if a query image with a panda is available, the system cannot precisely tell what the user seeks. The user might seek for an animal panda, a group of pandas, or a panda toy. Relevance feedback[1] provides the means for understanding the user's query concept better, by letting him or her give feedback regarding which images come closer to the query concept. But users might be unwilling to take the time for numerous iterations. If one begins with a Google text search, for example, it would be time-consuming to examine even just a small number of documents listed by the search engine. Fortunately for image retrieval, the time it takes to decide whether an image is relevant to a query concept is much shorter. Even so, we cannot always rely on users to provide a large number of relevant/irrelevant instances to use for training. Thus, a typical learning task for the classifier might be to infer a target concept with fewer than one hundred training instances, although the number of low-level features (the dimension of the *input space*) is usually much higher ($D > N$). (The distinction between *input space* and *feature space* is important. Input space refers to the space where the feature vectors reside. Feature space refers to the space onto which the feature vectors are projected via kernel methods [49].)

Semantic annotation (attaching text labels to images) suffers from training-data scarcity in a slightly different way. Although it might be technically feasible to label a large number of training images, doing so would not be economically practical. Given a fixed amount of budget and time to label N training instances, we would like the labeled pool to provide maximum information to enable a classifier to propagate annotation with maximal accuracy [4].

[1] Relevance feedback was first proposed in the information retrieval community for text [11], and was introduced to the image retrieval community by the works of [12,13].

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Both query-concept learning and annotation are problems related to supervised learning. Supervised learning embodies two approaches: *generative* and *discriminant*. The *generative* approach estimates the probability density function for each class (or concept), and then uses the density function to predict an unseen instance. The *discriminant* approach predicts the posterior probability of an unseen instance directly. In cognitive science, the most popular probabilistic model employed for the task of classification is Bayesian Networks (BNs) [14,15]. BNs are easy to interpret, if the model of a concept can be learned. However, it is well known that learning a BN model with even just a moderate number of features is an NP-hard problem [16]. In the $D > N$ scenario that image perception comes up against, learning a BN is simply impossible. (Stated in an analogical way, one cannot solve D variables with $N < D$ equations.)

The discriminant approach, on the other hand, can at least offer a solution even when $D > N$. In particular, the dual formulation of *Support Vector Machines* (SVMs) [17] consists of N free variables to solve, independent of the value of D. Even though the solution might overfit the data when $N > D$, it is the best that one can achieve under the circumstance. Practical experience tells us that if training instances are well chosen and representative, the concern of overfitting can be substantially mitigated.

To address the training-data scarcity ($D > N$) challenge, we consider three approaches. First, given N training instances from which to select, we choose the N most informative (most useful) instances for learning the target concept. Second, using the selected N instances, we infer additional training instances for increasing N. Third, we would employ dimensionality reduction methods for reducing D.

3.1 Approach #1. Making N Instances Most Useful

The first challenge of *query-concept learning* is to find some relevant objects so that the concept boundary (between images that are relevant and those that are irrelevant) can be fuzzily identified. For this discussion, we assume that keywords are not available to assist query processing. We would like to see how far statistical methods could go to learn a concept from image content (content refers to perceptual features like color, texture, and shape). Finding a relevant object can be difficult if only a small fraction of the dataset satisfies the target concept. For instance, suppose the number of desired objects in a one-million-image dataset is *1,000* (i.e., *0.1*% of the dataset). If we randomly select *20* objects per round for users to identify relevant objects, the probability of finding a positive sample after five rounds of random sampling is just *10*%---clearly not acceptable.

We can improve the odds with an intelligent sampling method like the *Maximizing Expected Generalization Algorithm* (MEGA) [6], which finds relevant instances quickly, to initialize *query-concept learning*. To explain without getting into mathematical details, MEGA uses given negative labeled instances to remove more negative instances. In addition, MEGA uses positive instances to eliminate irrelevant features. More specifically, MEGA models a query concept in k-CNF, which can express almost all practical query concepts; and MEGA uses k-DNF to model the sampling space. For instance, a 2-CNF of three features a, b, and c can be written as $(a \wedge b \wedge c \wedge (a \vee b) \wedge (a \vee b) \wedge (a \vee b))$; and a 2-DNF as $(a \vee b \vee c \vee (a \wedge b) \vee (a \wedge b) \vee (a \wedge b))$. A positively-labeled instance can eliminate terms in k-CNF, and hence makes it more general. A negatively-labeled instance can eliminate terms in k-DNF, and hence makes it more restrictive to remove additional negative instances. The two complementary steps---concept refinement and sample selection---increase the probability of finding relevant instances, even if only negative

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instances are available initially. To continue with the one-million-image dataset example, if we can reduce the search space from one million to thirty thousand by reducing negative instances, we simultaneously increase the odds of finding a positive instance from *0.1*% to *3.3*%. Thus, we increase from *10*% to *97*% the probability of finding one positive instance after five rounds of *20* random samples.

Once some relevant and some irrelevant instances have been identified, we can employ SVMActive [7] to refine the class boundary. Intuitively, SVMActive works by combining the following three ideas:

1. SVMActive regards the task of learning a target concept as one of learning an SVM binary classifier. An SVM captures the query concept by separating the relevant images from the irrelevant images with a hyperplane in a projected space (see Figure 4), usually a very high-dimensional one. The projected points on one side of the hyperplane are considered relevant to the query concept and the rest irrelevant.

2. SVMActive learns the classifier quickly via active learning. The active part of SVMActive selects the most informative labeled instances with which to train the SVM classifier. This step ensures fast convergence to the query concept in a small number of feedback rounds. The most informative instances are the ones in the margin nearest the hyperplane (the filled circles in Figure 4). These instances are most uncertain about their positive or negative membership, whereas the points farther away from the hyperplane on the positive (negative) side are more certain to be positive (negative).

3. Once the classifier is trained, SVMActive returns the top-k most relevant images. These are the k images farthest from the hyperplane on the query concept side.

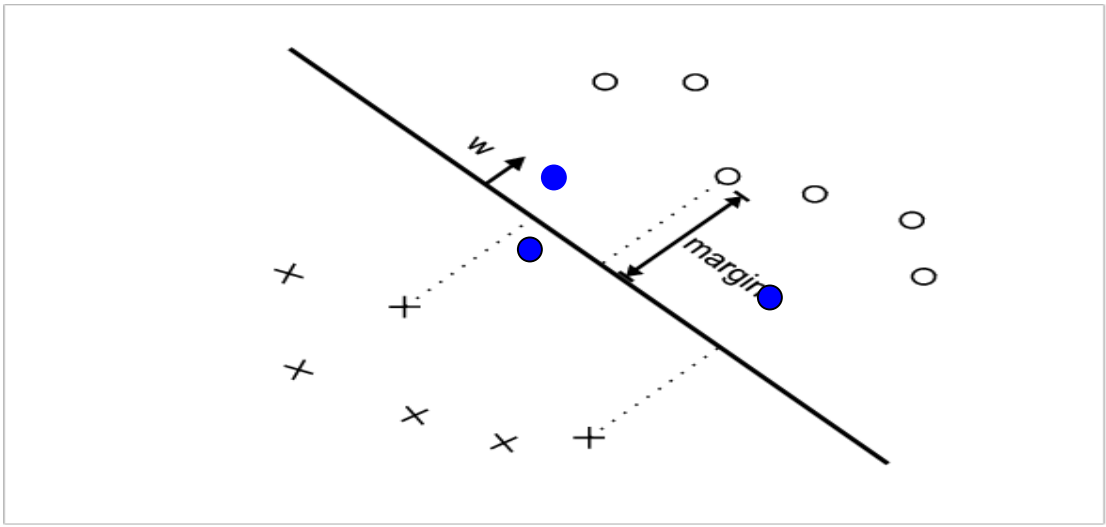


Figure 4. SVM Hyperplane, A Simple Illustrative Example.

In short, we can use MEGA to find initial relevant images, and then switch to SVMActive for refining the binary classifier and ranking returned images. This method can effectively learn a query concept when the *complexity* of a concept is not too high. We will define *concept complexity* in Section 4.2, and present remedies for searching concepts of high complexity.

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Remarks

R. Tibshirani remarks that when N < D, the employed statistical model should be simple (or the number of free variables should be small) so that the resulting solution would not overfit the data. The nearest shrunken centroids scheme [43] proposed by R. Tibshirani and T. Hastie shows promising results on analyzing DNA microarray data, where N < D. However, perceptual features interact with one another in high orders to depict an image concept [6]. Therefore, employing a kernel function to work with SVMs seems to be the logical choice. As previously mentioned, the dual formulation of SVMs consists of only N free variables (N α parameters), independent of D. Carefully selecting training data through active learning can alleviate concerns about overfitting.

3.2 Approach #2. Increasing N

The second approach is to increase the number of training instances from the given N instances [18]. As counter-intuitive as this may sound, the scarcity of negative training instances is more problematic than the scarcity of positive training instances for our learning algorithms. This is because describing a concept such as "tigers" may be challenging, but describing a negated concept (in this case the "non-tiger" concept) can require potentially infinite information. (Only a finite number of images is needed to adequately portray the tiger concept, but the number of non-tiger images is infinite.) Negative training instances are significantly easier to come by, but at the same time, the number of negative samples needed to depict a negated concept can be enormous. Thus, we need a substantial number of negative training instances to establish the class boundary accurately between negative and positive instances.

Let us revisit the SVMActive algorithm. At each round of sampling, it selects objects that are in the margin and also close to the dividing hyperplane. If the margin is wide (because of the lack of support at the negative side), the probability is low that a positive training instance can be found within that wide margin. Our goal is to find enough positive training instances to depict the target concept. However, the difficulty in gathering sufficient negative training instances slows down progress for finding positive training instances. The learning process is analogous to mining gold in a stream. The most productive way to harvest gold (relevant instances) is to quickly filter out water, mud and sand (negative-labeled instances) [18].

For solving the problem of insufficient training data, *transductive learning* has been proposed to work with different learning algorithms for various applications [19]. The basic idea of transduction is to leverage the unlabeled data near the labeled data. Suppose the nearest neighbors of negative training instances are always negative. We can then use the neighboring instances to substantially increase the negative-labeled pool. However, transduction must be performed with care, since the mis-labeling of data caused by wrong "guesses" can degrade the results. The performance of transduction is, unfortunately, consistently inconsistent. Its performance could be both application-dependent and dataset-dependent. For image retrieval, transduction may not be suitable for the following two reasons:

1. Nearest-neighbor sampling. Most image query-concepts are not convex, nor do they reside continuously in the input space. For instance, flowers of different colors tend to spread out in the color-feature dimension. Thus, to describe either a flower or a non-flower concept, we might need to *explore* the input space more aggressively. That is, we want to look for other possible flower (non-flower) images away from neighborhood of the already known flower

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(non-flower) images. Transductive learning, however, *exploits* only in the nearest neighborhood of the existing labeled instances. Suppose a picture of a lake (one of many non-flower instances) is labeled negative. Using the pictures neighboring this lake picture, which are likely to contain lakes or something visually similar, may not be very productive in refining the description of the negated concept (non-flower).

2. Noise factor. Mislabeled data due to noise can reduce the prediction accuracy of the classifier. In the usual case where relevant images are rare, performing transduction runs a very high risk of introducing false relevant instances, which thereby reduces class-prediction accuracy.

To avoid the aforementioned problems, we proposed employing *co-training* to increase N in a more meaningful and accurate way [18]. The problem of using a large unlabeled sample pool to boost performance of a learning algorithm is considered within the framework of co-training [20]. A broad definition of co-training is to provide each instance with multiple distinct views. Distinct views can be provided by different learning algorithms -- that is, by using MEGA and SVMActive at the same time. Information sharing between distinct views can increase especially the negative-labeled pool. Here, we propose another co-training method, which provides each training instance with distinct views, via subspace learning, to boost the pool of negative-labeled instances. As in *query-concept learning* scenario, where the percentage of positive-labeled instances is rare, boosting the positive pool directly may not be productive. Thus, we attempt to boost the negative pool to increase the probability of finding positive instances. This method recursively conducts subspace co-training at each feedback iteration in the following steps:

1. Divide the input space into G subspaces.
2. Conduct parallel training in these G subspaces using labeled training dataset L.
3. Use the G resulting learners to label the unlabeled pool and yield a new set of labeled instances L'.
4. Replace L with L'.
5. Go back to Step 1 until no more labeled instances can be inferred (that is, until $L' = \varnothing$).

3.3 Approach #3. Reducing D

When D is large, the training instances in the high-dimensional input space become very sparse (even when $N > D$). The sparsely populated space causes two theoretical obstacles for conducting statistical inferences. First, given a query instance, its nearest neighbors tend to be equally distanced when data are uniformly distributed [21]. (Since most image datasets are not uniformly distributed in the input space, this concern may not be applicable to our case.) Second, the nearest neighbors are not "local" to the query point. This second mathematical problem makes class prediction difficult.

Many methods have been proposed for performing dimensionality reduction [9], such as principal component analysis (PCA) and independent component analysis (ICA). Recently, several dimensionality-reduction algorithms have been proposed to find nonlinear manifolds embedded in a high-dimensional space. Among the proposed methods, Isomap [22], local linear embedding (LLE) [23], and kernel PCA (KPCA) [24] have been applied to tasks of image clustering and image retrieval [25]. However, the scenarios in which manifold learning appears to be effective seem rather contrived. For instance, the widely used Swiss-roll example [23] is a

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three-dimensional structure on which data are densely populated. Several examples of face and object images presented in the literatures change their poses only slightly from one image to another, so manifolds can easily be discovered. Moreover, none of the demonstrated scenarios have considered noise as a factor to seriously challenge manifold learning.

The major drawback of these dimension-reduction methods in the context of *query-concept learning* is that they reduce dimensionality in a universal mode with respect to the entire dataset, and with respect to all users. Reducing dimensionality in such a one-size-fits-all way entirely disregards individuals' subjectivity. Our work [26,27] shows that a perceptual distance function is not only *partial* in dimensionality, but also *dynamic* in the subspace where the similarity between two images is measured. We thus propose using *Dynamic Partial Function* (DPF) to perform adaptive dimensionality reduction.

Let us briefly summarize why DPF works. Some important work in the cognitive psychology community has provided ample evidence to show that not all features need to be considered in measuring similarity between a pair of objects (images) [28,29]. Most recently, Goldstone [30] shows that similarity perception is the *process* that determines the respects (features or attributes) for measuring similarity. More precisely, a similarity function for measuring a pair of objects is formulated only *after* the objects are compared, not *before* the comparison is made; and the respects for the comparison are *dynamically* activated in this formulation process. Let us use a simple example to explain. Suppose we are asked to name two places that are similar to England. Among several possibilities, "Scotland and New England" could be a reasonable answer. However, the respects in which England is similar to Scotland differ from those in which England is similar to New England. If we use the shared attributes of England and Scotland to compare England and New England, the latter pair might not seem similar, and vice versa. Thus, a distance function using a fixed set of respects (such as the traditional weighted Minkowski function) cannot capture objects that are similar in different sets of respects. (Please refer to [27] for further information about how DPF dynamically select a subset of features in a pair-dependent way.)

A seamless way to integrate this partial and dynamic dimensionality reduction method into *query-concept learning* is to introduce a new DPF kernel function as

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(\Sigma\, j \in \Delta m \;\; |\mathbf{x}j - \mathbf{x}'j|^2 / 2\,\sigma^2\right),$$

where Δm is the set that contains m out of D dimensions that have the smallest feature differences between instances **x** and **x'**, the j suffix denotes the jth feature of image **x** and **x'**.

4. Imbalance of Training Classes

A subtle but serious problem that hinders the process of classification is the skewed class boundary caused by imbalanced training data. To illustrate this problem, we use a 2D checkerboard example. Figure 5 shows a checkerboard of 200×200 square divided into four quadrants. The top-left and bottom-right quadrants are occupied by negative (majority) instances, but the top-right and bottom-left quadrants contain only positive (minority) instances. The lines between the classes are the "ideal" boundary that separates the two classes. In the rest of this section, we will use *positive* when referring to minority instances, and *negative* when referring to majority instances.

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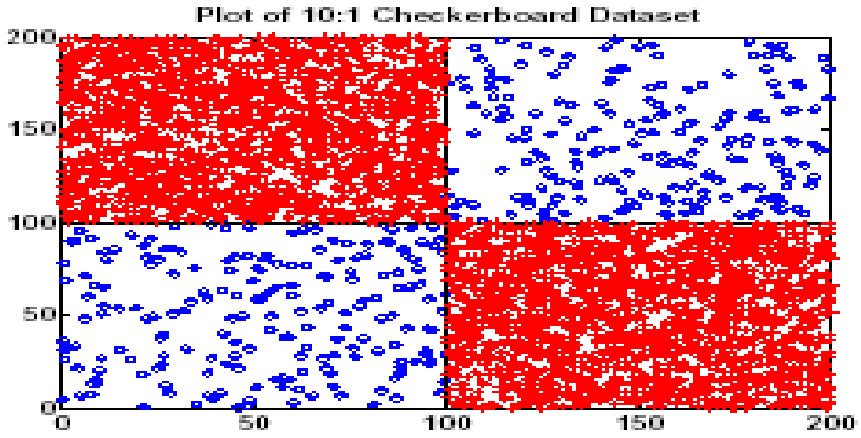


Figure 5. Two-class Checker Board Setup

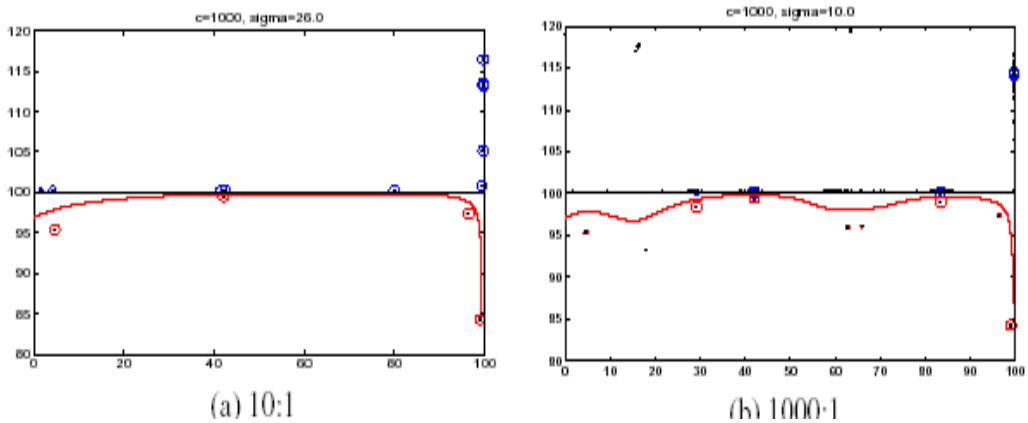


Figure 6. Biased Boundary

Using SVMs as the classifier, we plot in Figures 6(a) and (b) the boundary distortion between the two left quadrants of the checkerboard under two different negative/positive training-data ratios. Figure 6(a) shows the SVM class boundary when the ratio of the number of negative instances (in the quadrant above) to the number of positive instances (in the quadrant below) is *10:1*. Figure 6(b) shows the boundary when the ratio increases to *10,000:1*. The boundary in Figure 6(b) is much more skewed toward the positive quadrant than the boundary in Figure 6(a), which results in a higher incidence of false negatives.

Both *query-concept learning* and *semantic annotation* may suffer from the imbalanced training-class problem, since the target class is always overwhelmingly outnumbered by the other classes (that is, the relevant class is often rare and outnumbered by the irrelevant class). We consider two approaches for remedying the imbalanced training-data problem for SVMs: the algorithmic approach and the data-processing approach. The algorithmic approach focuses on modifying the kernel function or the kernel matrix, and the data-processing approach focuses on processing the training data.

4.1 Approach #1. Algorithmic Remedy

Examining the class prediction function of SVMs

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$$\text{sgn}(f(\mathbf{x}) = \Sigma i\ y_i\ \alpha_i\ K(\mathbf{x},\mathbf{x}_i) + b,\ i = 1,\ldots n$$

we can identify three parameters that affect the decision outcome: b, αi, and K. In [31,32], we show that the only effective algorithmic method for tackling the problem of imbalanced training classes is through modifying the kernel K or the kernel matrix[2] $\mathbf{K}_{i,j}$ adaptively to the training data distribution. We propose using adaptive conformal transformation to dynamically enlarge (or scale) the spatial resolution of minority and majority areas around the separating boundary surface in the feature space, such that the SVM margin could be increased while the boundary becomes less skewed. Conformal transformation modifies the kernel function by changing the Riemannian geometrical structure induced by the mapping function. For data that do not have a vector-space representation (such as video sequence data), we apply transformation directly on the kernel matrix. Our experimental results on both UCI and real-world image/video datasets show the proposed algorithm to be very effective in correcting the skewed boundary. For further details, see [31,32].

4.2 Approach #2. Data Processing Remedy

The *bagging* process subsamples training data into a number of *bags*, trains each bag, and aggregates the decisions of the bags to make final class predictions. We propose *quasi-bagging*, which performs subsampling somewhat differently. For each bag, quasi-bagging subsamples only the majority class, but takes the entire minority class. By subsampling just the majority class, we improve class balance in each training bag. At the same time, using a large number of bags reduces the variance caused by the subsampling process. Please refer to [1] for details.

5. Synergistic Context and Content Fusion

So far, we have addressed the problem of query-concept learning: how to find the right combination of perceptual features to depict a concept. We have not discussed how keywords can be annotated to images to provide text-based searches. Using perceptual features alone, certain semantics cannot be depicted (perceptual features cannot reliably tell e.g., where, when, or who). Keyword annotation can be complementary to perceptual features. In this section, we broaden the scope of our discussion by bringing keyword annotation into the "picture." We first discuss how keywords can be semi-automatically provided for images; we then discuss how content and context (defined shortly) can be effectively combined to improve image organization and retrieval quality.

5.1 Applying Keywords to Images

Traditional keyword annotation has been performed either by people manually, or by statistical methods that propagate annotation from annotated images to unannotated ones via image content. Although progress has been made in semi-automatic annotation [2-5], the achieved quality is far from satisfactory. At least a couple of obstacles make image annotation difficult. First, images are taken under varying environmental conditions (e.g., lighting, movements, etc.)

[2] Given a kernel function K and a set of instances $S = \{\mathbf{x}_i, y_i\}$, i=1,…,n, the kernel matrix (Gram matrix) is the matrix of all possible inner-products of pairs from S, $\mathbf{K}_{i,j} = K(\mathbf{x}_i, \mathbf{x}_j)$.

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and with varied resolutions. The same Eiffel tower pictures taken by two different models of cameras, from two different angles, and at different times of day seem different enough to be mislabeled. Thus, keyword propagation via image content can be unreliable. Second, although some progress has been made [50], reliable object segmentation is not attainable in the foreseeable future [3]. When we are able to segment objects with high accuracy, recognizing objects will be easier.

Fortunately, recent hardware advances are beginning to assist. With advances in lenses on camera phones, many pundits believe that camera phones will replace digital and film cameras within a few years. A camera with the ability to record time, location, and voice, and to provide camera parameters (which can tell the environmental characteristics at the photo-taking time) offers renewal of optimism for achieving high-quality automatic annotation.

Semantic labels can be roughly divided into two categories: *wh* labels and *non-wh* labels. Wh-semantics include time (*when*), people (*who*), location (*where*), landmarks (*what*), and event (inferred from *when*, *who*, *where*, and *what*). Providing the *when* and *where* information is trivial. Already cameras can provide time, and we can easily infer location from GPS or CellID. However, determining the *what* and *who* requires contextual information in addition to time, location, and photo content. More precisely, contextual information provided by cameras includes *time*, *location*, *camera parameters*, *user profile, and voice*. Content of images consists of perceptual features, which can be categorized into *holistic features* (color, shape and texture characteristics of an image), and *local features* (edges and salient points of regions or objects in an image). Besides context and content, another important source of information (which has been largely ignored) is the relationships between semantic labels (which we refer to as *semantic ontology*). To explain the importance of having a semantic ontology, let us consider an example with two semantic labels: *outdoor* and *sunset*. When considering contextual information alone, we may be able to infer the *outdoor* label from camera parameters: focal length and lighting condition [33]. We can infer *sunset* from time and location [34]. Notice that inferring *outdoor* and *sunset* do not rely on any common contextual modality. However, we can say that a *sunset* photo is *outdoor* with certainty (but not the other way). By considering semantic relationships between labels, photo annotation can take advantage of contextual information in a "transitive" way.

To fuse *context*, *content*, and *semantic ontology* in a synergistic way, we propose EXTENT [35], an inferencing framework to generate semantic labels for photos. EXTENT uses an influence diagram [36] to conduct semantic inferencing (see details in [s]). The variables on the diagram can either be *decision variables* (i.e., causes), or *chance variables* or (i.e., effects). For image annotation, decision variables include time, location, user profile, and camera parameters. Chance variables are semantic labels. However, some variables may play both roles. For instance, time can affect some camera parameters (such as exposure time and flash on/off), and hence these camera parameters are both decision and chance variables. Finally, the influence diagram connects decision variables with chance variables with arcs weighted by *causal strength*.

To construct an influence diagram, we rely on both domain knowledge and data. In general, learning such a probabilistic graphical model from data is an NP hard problem [16]. For image annotation, however, we have abundant prior knowledge about the relationships between context, content, and semantic labels, and we can use them to substantially reduce the hypothesis space to search for the right model. For instance, time, location, and user profile, are

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independent of each other (no arc exists between them in the diagram). Camera parameters such as exposure time and flash on/off depend on time, but are independent of other modalities (hence some fixed arcs can be determined). The semantic ontology provides us the relationships between words. The only causal relationships that we must learn from data are those between context/content and semantic labels (and their causal strengths).

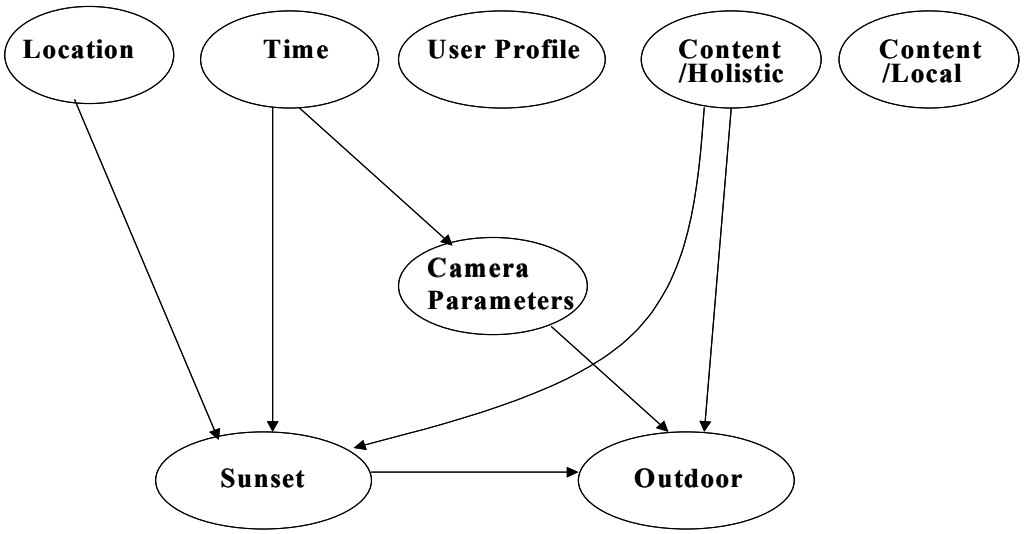


Figure 7. Causal Diagram

Let us use a model generated by our algorithm to illustrate the power of our fusion model. Figure 7 shows the learned diagram for two semantic labels: *outdoor* and *sunset*. In addition to the arcs that show the causal relationships between the variables, and between the variables and the labels, we also see that the relationship between outdoor and sunset can also be encoded.

Case Study - Landmark Recognition

We present a case study on landmark recognition using the influence diagram. Recognizing objects (landmarks or people) in a photo remains a very challenging computer-vision research problem. However, with available contextual information, such as time, location, and a person's social network, recognizing objects among a much-limited set of candidates is not such a daunting task. For example, given the photo-taking location as downtown Santa Barbara, the interesting landmarks in the photo are limited. Given a birthday party of a person, the attendees are most likely to be his/her friends and family. With candidate objects being limited, matching becomes easier, and matching time becomes shorter, so we can afford to employ more expensive matching techniques for improved accuracy.

Some recent works [34,37] annotate digital images using the metadata of spatial and temporal context. Nevertheless, these methods suffer from several drawbacks. In [34,37], GPS devices are used to infer the location where a photo is taken. However, the objects in a photo may be far away from the camera, not at the GPS coordinate. For instance, one can take a picture of the Bay Bridge from many different locations in the cities of Berkeley, Oakland, or San Francisco. The GPS information of the camera cannot definitely infer landmarks in a photo. Second, a landmark

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in a photo might be occluded and hence not deemed important. For instance, a ship might occlude a bridge; a person might occlude a building, etc. Also, a person can be inside or outside of a landmark. A robust system must analyze content to identify the landmarks precisely. The work of [34] uses temporal context to find time-of-day, weather, and season information. These additional metadata provide only a few drops in the bucket to quench thirst, and the proposed method cannot scale up for annotating vast quantities of other semantics.

For landmark recognition, we use two modalities: *location* and *local features*. We select these two modalities to construct our influence diagram based on our domain knowledge. Although other metadata might also play a role in determining a landmark, we keep this case study simple. For local features, we follow the SIFT-line of work proposed by [38,39]. (An overview of SIFT was presented in Section 2.)

Our testbed (referred to hereafter as the Stanford dataset) was obtained from Mor Naaman [34]. The dataset was constructed by collecting photographs taken by visitors to the Stanford Visitor Center. All photographs were taken on the Stanford campus and are annotated with GPS information. From this dataset, we used a subset containing about *1,000* images, and added *60* pictures taken by the author of this article. To evaluate the EXTENT system, we selected photos with GPS coordinates around the Memorial Church and Hoover Tower (two important landmarks on the Stanford campus). All images were rescaled to *320×240*, before SIFT feature extraction was performed.



Figure 8. Landmark Queries

Each image in the dataset was individually processed and matched with sample images that contain candidate landmarks. We used a distance threshold to separate likely matches from non-matches. If the distance (computed using the SIFT features) between the query image and a landmark sample came within the threshold, the landmark would be a possible match. If no possible match was found (after comparing with all landmark samples), we concluded that the image contained no landmark. Otherwise, the best match was used to annotate the image.

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We made three queries on the dataset: "Memory Church," "Hoover Tower," and "Rodin Garden" (Figure 8). Each query has *10* images containing the landmark taken at different time, from different angles, in the dataset. For each query, we used one of the ten images as the query, and the rest nine were mingled in the *1,000*-image dataset. We performed this leave-one-out query for each of the ten images, and for three landmarks. Using GPS alone gives us just *33.3*% accuracy. This is because we have three landmarks in front of the Stanford Quad, and the information about Quad cannot give us a specific landmark, and we have to resort to a random guess. Using just SIFT features also produces a disappointing result of *60*%. This is because too many Stanford buildings exhibit the same textures and shapes of the Hoover Tower. There are also a couple of locations displaying Rodin sculptures. Finally, combining GPS and SIFT gives us *95*% accuracy, a much better result, thanks to the synergistic fusion of the two modalities.

5.2 Keyword Assisted Concept-dependent Active Learning

With keywords being available, the remaining question is how to use them effectively to assist image retrieval. One straightforward way is to turn image retrieval into text retrieval. However, subjective annotation and multiple senses of words prevent this approach from being invariably effective. Furthermore, even if the semantics of an image matches a query keyword, the user may prefer certain visual aspects that words cannot describe. Therefore, perceptual features must also be considered in order to offer a complete solution.

We present a *concept-dependent* active learning algorithm [40], which uses keywords and perceptual features in a synergistic way. Concept-dependent active learning (CDAL) aims to solve one key problem: concept aliasing. Revisit the active learning algorithm presented in Section 2.1. A major shortcoming of the method is that when a query-concept's *complexity* is high, active learning may not be able to find sufficient images relevant to the concept. This is because the concept can be cluttered with others in the perceptual space. More specifically, we formally define *concept complexity* as the level of difficulty in learning a target concept. To model concept complexity, we use two quantitative measures [41]: *scarcity* and *isolation*.

Scarcity measures how well represented a concept is in the retrieval system. We use *hit-rate*, defined as the percentage of data matching the concept, to indicate scarcity. As we assume that each keyword is equivalent to a concept, the hit-rate of a keyword is the number of images being annotated with that keyword. This parameter is dataset-dependent; while a concept such as *outdoor* is very general and may produce a high hit-rate, other general concepts such as *eagles* may be scarce simply because the system does not contain many matching images. Similarly, a very specific concept such as *laboratory coat* could have a high hit-rate solely because the system has many such images.

Isolation characterizes a concept's degree of separation from the other concepts. We measure two types of isolation: *input space* isolation and *keyword* isolation. The input space isolation is considered low (or poor) when the concept is commingled with others in the perceptual space. When the keywords used to describe a concept has several meanings or senses, the keyword isolation is considered poor; very precise keywords provide good isolation. An example of a poorly isolated keyword is "feline," which is often used to annotate images of tigers, lions and domestic cats. If the user seeds a query with "feline," it is difficult to tell which sense of the word the user has in mind. A well-isolated seed keyword like "Eiffel Tower" has less ambiguity; we can be quite sure that the user is thinking about the famous Parisian landmark.

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We describe how scarcity and isolation can be quantified in [40,41]. Once the complexity of a concept can be measured, we can adjust SVMActive in a concept-dependent way. The key of CDAL is to reduce the complexity of the concept, so that the learnability of the target concept is improved.

CDAL consists of a state-transition table and two major algorithms: *disambiguate keywords* (DK) and *disambiguate input-space* (DS). First, CDAL addresses the scarcity problem by using keywords to seed a query. Thus, we eliminate the need to search the entire dataset for a positive image. The user can type in a keyword (or keywords) to describe the target concept. Images that are annotated with that keyword are added to the initial unlabeled pool *U*. If the number of images with matching keywords is small, we can perform query expansion using a thesaurus to obtain related words (synonyms) that have matching images in our dataset. For each related semantic, we select matching images and add them to *U*.

Using *U* to conduct learning rather than using the entire dataset improves the *hit-rate*. At this juncture, the learnability state can be in any of the four states depicted in Figure 9. The rows indicate whether the specified keyword enjoys good isolation or suffer from poor isolation. The columns show that the query concept may be well or poorly isolated in the input space formed by the perceptual features. The goal of CDAL is to move the learnability state of the target concept to the ideal state A, where both keywords and perceptual features enjoy good isolation. In the remainder of this section, we sketch how CDAL works. For further details, please refer to [40].

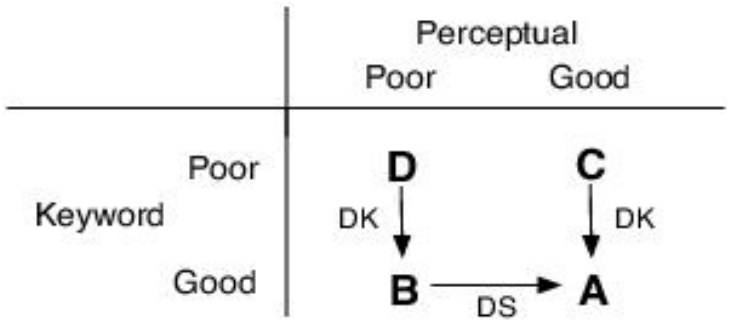


Figure 9. CDAL States and State Transitions

State C - Keyword Disambiguation

In state C, the keyword isolation is poor (due to aliasing). Thus, we first need to disambiguate the keywords by presenting images of various semantics to the user for selection. Once the semantic is understood, the learnability makes a transition to state A.

Algorithm DK helps state C to arrive at A. The algorithm first performs a lookup in the association table H (constructed through association mining [42] on keywords of images) for the query keyword *w*. If 2-itemsets are found, the algorithm randomly selects from *U* images that are annotated by the co-occurred word set. The resulting sample set *S* is presented to the user to

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solicit feedback. Once feedback has been received, CDAL removes from *U* images labeled with the words associated with the irrelevant images. Notice a subtle but important difference between our query-expansion approach and the traditional method that uses a semantic ontology to find synonyms. For the task of image annotation (discussed in Section 4.1) a semantic ontology is helpful. For image retrieval, however, semantic ontology may be counter-productive. This is because images labeled by a keyword's synonyms might not exist in the dataset. Using co-occurrence, mined by association rules on the dataset, provides information about the dataset, and hence is useful for disambiguating keyword semantics in a dataset-dependent way.

State B - Input-space Disambiguation

State B has poor perceptual isolation but good keyword isolation. If the number of matching images yielded by the good keyword is small, we can perform query expansion to enlarge the unlabeled sample pool. For these poorly isolated concepts, we attempt to improve the perceptual isolation of the sample pool through input-space disambiguation. CDAL employs DS to achieve this improvement. The algorithm removes from *U* the instances that can interfere with the learnability of the target concept.

Given the current labeled pool of images, DS first generates two word lists: *P* from positive-labeled images and *N* from negative-labeled images. In the third step, DS checks the annotation of each image in the unlabeled pool for matches to words in *P* and *N*. If there are more negative keyword matches, CDAL removes that image from *U*. Once perceptual isolation is enhanced, state B makes the transition to A.

State D - Keyword & Space Disambiguation

State D suffers from both poor semantic and poor input-space isolation. To remedy the problem, CDAL first improves the keyword isolation using the same DK strategy as in state C. Thereafter, the state is moved to state B, and the perceptual isolation is improved by applying state B's method for disambiguation in the input space. Figure 8 shows the path that D takes to reach the ideal state A.

6. Concluding Remarks

We have discussed in this article several statistical models and algorithms for addressing challenges of image perception. Essentially, these models and algorithms attempt to model human perception of images, and thereby achieve more intuitive and effective image annotation and retrieval. We enumerated three major technical challenges: scarcity of training instances, imbalanced training data, and fusion of context and content. These challenges run parallel to the challenges human perception comes up against. For scarcity of training instances, most of us learned in our childhood to recognize, for example, an animal by seeing just few example pictures. For imbalanced training data, information relevant to each concept we attempt to learn plays a minority role in our knowledge base and experience. For context and content fusion, we seem to be at ease to correctly interpret an expression in different contexts. We showed the mathematical difficulties in dealing with these challenges, and presented our solutions. With our integrated approach, combining keyword with perceptual features and combining context with content, we have built systems that can perform at a satisfactory level and can be used in practical scenarios.

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